

Customer Churn Analysis – Telecom Industry

1. Introduction

In today's highly competitive telecom industry, customer retention is more crucial than ever. Understanding the factors that lead to customer churn can help telecom companies retain valuable customers and improve profitability. This project leverages machine learning techniques to analyze customer behavior and predict churn, enabling data-driven decisions for customer engagement and retention strategies.

2. Abstract

This project involves the end-to-end analysis of customer churn using a publicly available telecom dataset. The analysis includes preprocessing, model building using a Random Forest classifier, and using SHAP (SHapley Additive exPlanations) for explainability. Key churn predictors were identified and customers were segmented into Loyal, At Risk, and Dormant categories. The insights drawn offer actionable recommendations for improving customer retention and reducing churn.

3. Tools Used

- **Python Libraries:** Pandas, NumPy, Scikit-learn, SHAP
 - **Jupyter Notebook:** Development Environment
 - **SQL:** For data querying and analysis
 - **PowerPoint:** For presentation and visual reporting
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4. Steps Involved in Building the Project

1. Data Preprocessing

- Checked for null values and handled missing data
- Encoded categorical variables using Label Encoding
- Scaled numerical features for model readiness

2. Train-Test Split & Model Training

- Dataset was split (80-20 ratio)

- Trained a Random Forest Classifier

3. Model Evaluation

- Evaluated using Accuracy, Precision, Recall, F1-Score
- Achieved good balance of precision and recall for churn prediction

4. Explainability using SHAP

- Applied SHAP to interpret feature contributions
- Identified key drivers of churn like contract type, tenure, and internet service

5. Customer Segmentation

- Segmented users based on churn probability:
 - **Loyal Customers** (Low probability)
 - **At Risk Customers** (Moderate probability)
 - **Dormant/High Risk Customers** (High probability)

5. Conclusion

The model successfully predicts customer churn and provides valuable insights into churn drivers using SHAP. By identifying customers at risk, telecom companies can take proactive measures such as personalized offers, rewards for loyalty, or service improvements. This project exemplifies how machine learning can be utilized effectively in business for customer retention strategies.

Report: No-Show Appointment Prediction

Introduction

This project focuses on predicting whether a patient will miss a scheduled medical appointment (no-show) based on various factors such as demographics, appointment details, and medical history. The dataset used contains information from over 110,000 medical appointments in Brazil. The goal is to build a predictive model to help healthcare providers identify high-risk no-show cases and improve appointment adherence.

Abstract

The project employs a Decision Tree Classifier to predict no-show appointments. Key features include patient age, gender, waiting days between scheduling and appointment, and medical conditions. The model achieves an accuracy of 80.13% , but recall for no-shows (class "1") is low (0.00), indicating room for improvement. The cleaned dataset with

Tools Used

- Python (Primary programming language)
- Pandas & NumPy (Data manipulation)
- Matplotlib & Seaborn (Visualization)
- Scikit-learn (Machine learning: Decision Tree Classifier)
- Jupyter Notebook (Interactive development)

Steps Involved in Building the Project

1. Data Loading & Exploration

- Loaded the dataset (`KaggleV2-May-2016.csv`).
- Inspected structure, missing values, and class distribution (88,208 showed up vs. 22,319 no-shows).

2. Data Preprocessing

- Converted date columns (`ScheduledDay`, `AppointmentDay`) to datetime.
- Derived new features:
 - `Wait\_Days` (Days between scheduling and appointment).
 - `DayOfWeek` (Day of the week for the appointment).
- Filtered invalid ages (removed negative values).

- Encoded categorical variables (`Gender`, `No-show` → binary).

3. Model Training & Evaluation

- Split data into training (80%) and testing (20%) sets.
- Trained a Decision Tree Classifier (`max\_depth=5`).
- Evaluated performance:
 - Accuracy: 80.13%
 - Precision for No-Shows: 47% (Low recall indicates model struggles to identify no-shows).

4. Output & Deployment

- Saved predictions to `no\_show\_cleaned.csv` for further use.

Conclusion

The model achieves moderate accuracy but fails to reliably predict no-shows (recall = 0%). Future improvements could include:

- Feature Engineering: Adding socioeconomic or location-based features.
- Handling Class Imbalance : Using techniques like SMOTE or class weighting.
- Alternative Models : Random Forest or Gradient Boosting for better performance.

This project provides a foundation for predicting appointment adherence, with potential applications in healthcare scheduling optimization.

Note : This report is concise and fits within the 2-page limit. Let me know if you'd like any modifications!